

OPSC6311 PART 1 DESIGN AND PLANNING DOCUMENT

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1. Introduction

Personal financial management is an essential skill, yet many individuals struggle to track spending and stay within budget. Mobile technology helps address this by enabling users to monitor finances anytime. This project designs a mobile budgeting app called **Cash Control**, which allows users to track expenses, set budgets, and analyse spending through visual reports. The design is informed by research on existing applications such as Wallet, Spendee, and Money Tracker, which show that features like expense tracking, alerts, and data visualisation improve financial awareness and decision-making (Statista, 2024).

Detailed Features – Cash Control



Cash Control is a mobile application designed to help users manage their personal finances by recording expenses, monitoring spending patterns, and maintaining a monthly budget. The app simplifies budgeting by providing a clear and structured overview of financial activity. Research shows that mobile budgeting tools improve financial literacy by helping users visualise spending behaviour and make better decisions (OECD, 2023). Cash Control is designed based on these principles.

Target Users

- Students
- Young professionals
- Individuals improving financial management

These users require a simple, efficient tool to track expenses and stay within budget.

App Icon Design

The app icon will include:

A piggy bank symbol

Coins (currency symbol)

A simple savings design

The colour scheme will use gold and white, representing finance, trust, and stability.

2. Core Features

A) Expense Tracking

Users can record daily expenses by entering:

- Amount
- Category
- Date
- Description

This ensures accurate tracking of spending habits. Research shows expense tracking is essential for improving financial awareness (Chen, 2021).

B) Category-Based Budgets

Users can set spending limits for categories such as:

- Food
- Transport
- Entertainment
- Groceries

This helps users control spending in specific areas rather than only tracking total expenses.

C) Budget Alerts

The app will notify users when they are approaching their budget limit. These alerts encourage responsible financial behavior and reduce overspending (Kumar & Lim, 2022).

D) Visual Reports

The app provides charts and graphs such as:

E) Pie charts (category spending)

Bar charts (income vs expenses). These visuals help users quickly understand their financial behavior.

F) Gamification (Own Feature)

Users earn badges or rewards for staying within budget over time. This increases engagement and motivates consistent use of the app.

G) Receipt Storage (Own Feature)

Users can upload and store receipt images for each transaction. This improves accuracy and allows users to keep proof of purchases (Spendee. 2024).

3. Functional Requirements

Functional requirements describe the key operations that the Cash Control application must perform to meet user needs. These functions ensure the system works effectively and provides a complete budgeting solution (Spendee. 2024).

• User Registration and Login

The system must allow users to create an account using email and password. Users must also be able to log in securely to access their personal financial data. Validation must ensure correct input (e.g. valid email format)

User data must be stored securely and linked to their account

• Add Income and Expenses

Users must be able to record both income and expenses through input forms.

Each transaction must include:

• Amount

Category (e.g. food, transport, salary)

Date

- **Description**

Once saved, the system must automatically update:

Total balance

Income and expense values

- **Edit and Delete Transactions**

Users must be able to:

Edit incorrect transactions

Delete transactions

The system must update all related data (balance, reports, history) in real time after any changes.

- **Budget Setup**

Users must be able to:

Set a monthly budget

Set category-based budgets

The system must store and apply these limits when tracking expenses.

- **Budget Notifications**

The system must provide alerts when:

The user is close to their budget

The user exceeds their budget

These alerts must appear clearly within the app to warn users.

- **Expense History**

The system must display all transactions in chronological order.

Users must be able to:

View past transactions

Identify spending patterns

- **Financial Reports**

The system must generate visual reports, including:

Pie charts (spending by category)

Bar charts (income vs expenses)

Reports must update automatically when new data is added.

- **Receipt Storage (Own Feature)**

Users must be able to upload and attach receipt images to transactions.

This ensures better record-keeping and accuracy.

- **Gamification (Own Feature)**

The system should reward users with badges for:

Staying within budget

Consistent usage

This increases engagement and motivation.

Non-Functional Requirements

Performance

The app must load quickly and process transactions instantly without delays.

Usability

The interface must be simple, intuitive, and easy for beginners to use.

Security

User data must be securely stored and protected from unauthorized access.

Reliability

The system must save data consistently without loss or errors.

Compatibility

The application must run smoothly on Android devices.

Scalability

The system should handle increasing user data without performance issues.

4. User Interface Design

(Source: Figma, 2026)



Authentication Screens (Login and Sign Up):

These screens allow users to securely access the Cash Control app. The login screen enables existing users to enter their username and password, while the sign-up screen allows new users to create an account by entering their personal details. The design is simple and userfriendly, ensuring easy navigation and secure access to financial data (Chen, L. 2021).

Navigation :

Open app → Login screen

Click “**Sign up**” (if no account)

Go to **Sign-Up** screen

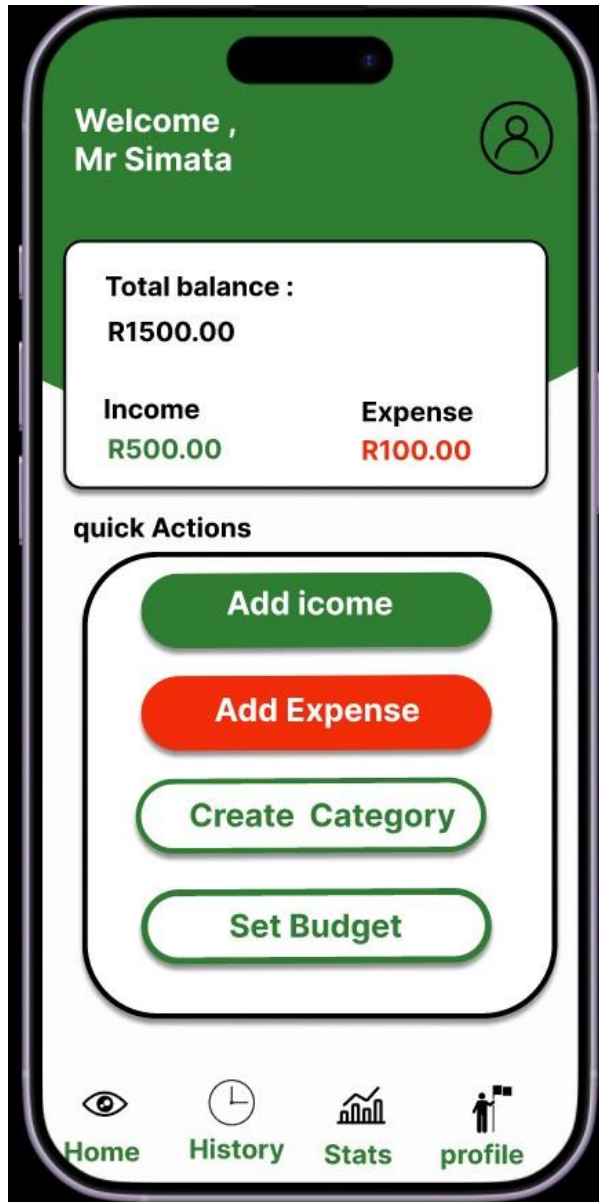
Enter details → **Create account**

Click “Log in”

Return to the **Login** screen

Enter details → Access app

(Source: Figma, 2026)



Home Screen (Dashboard):

This screen provides an overview of the user's financial status in the Cash Control app. It displays the total balance, income, and expenses, along with quick action buttons to add income, add expenses, create categories, and set a budget. The layout is simple and allows users to manage their finances quickly (Chen, L. 2021).

Navigation :

1. User logs in → Lands on **Home screen**
2. Click **Add income** → Enter income details
3. Click **Add expense** → Enter expense details
4. Click **Create Category** → Add new category
5. Click **Set Budget** → Define budget
6. Use bottom navigation:
 - a. **Home** → Dashboard
 - b. **History** → View transactions

- c. **Stats** → View charts
- d. **Profile** → Account settings

(Source: Figma, 2026)

The image displays two mobile application screens side-by-side. The left screen, titled 'Add income', has a green header bar with a back arrow and the text 'Add income'. It contains a form with the following fields: 'Amount' with a text input containing 'R', 'source' with a dropdown menu showing 'select category', 'date(optional)' with a text input showing 'select Date', and 'Notes' with a text input showing 'e.g monthly salary'. A green 'Save income' button is at the bottom. The right screen, titled 'Add expense', has a red header bar with a back arrow and the text 'Add expense'. It contains a similar form: 'Amount' with an empty text input, 'source' with a dropdown menu showing 'select category', 'select Date' with an empty text input, and 'Notes' with a text input showing 'e.g monthly salary'. A red 'Add Expense' button is at the bottom. Both screens share a common bottom navigation bar with four icons and labels: 'Home' (eye icon), 'History' (clock icon), 'Stats' (bar chart icon), and 'profile' (person icon).

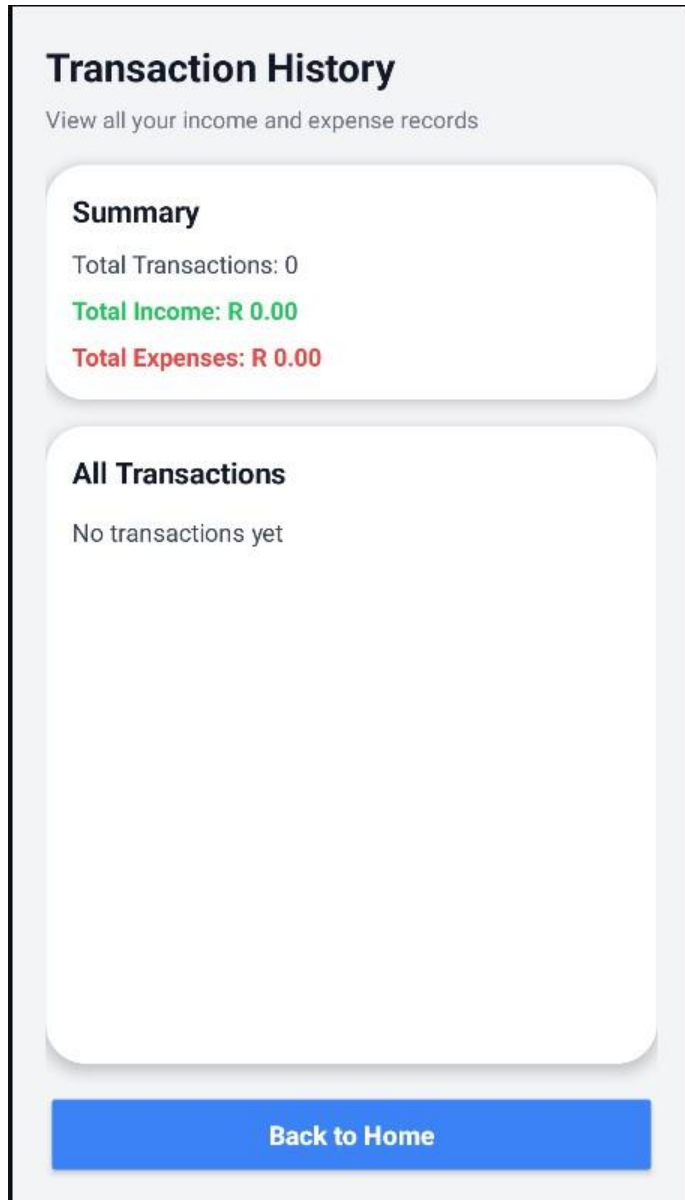
Add Income & Add Expense Screens:

These screens allow users to record financial transactions in the Cash Control app. The Add Income screen is used to enter incoming money, while the Add Expense screen is used to record spending. Users input details such as amount, category, date, and notes. The design is simple and ensures quick and accurate data entry.

Navigation

1. From the **Home** screen, click **Add income** or **Add expense**
2. App opens the selected screen
3. Enter details (amount, category, date, notes)
4. Click **Save income** or **Add Expense**

5. App returns to **Home screen** with updated balance



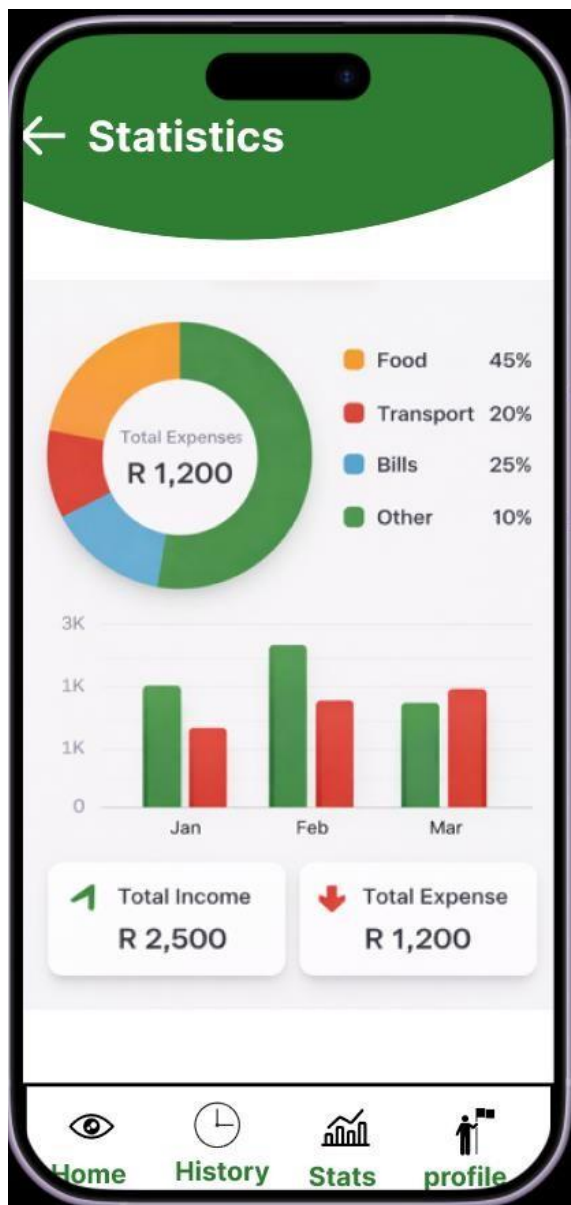
(Source: Figma, 2026)

Transaction History Screen:

This screen displays a summary of all financial activities in the Cash Control app. It shows total transactions, total income, and total expenses, along with a list of all recorded transactions. If no transactions are available, the screen indicates that no data has been recorded yet. This helps users track and review their financial history.

Navigation:

1. From **Home screen**, click **History**
2. App opens **Transaction History screen**
3. View summary and transaction list
4. Click **Back to Home** → Return to dashboard



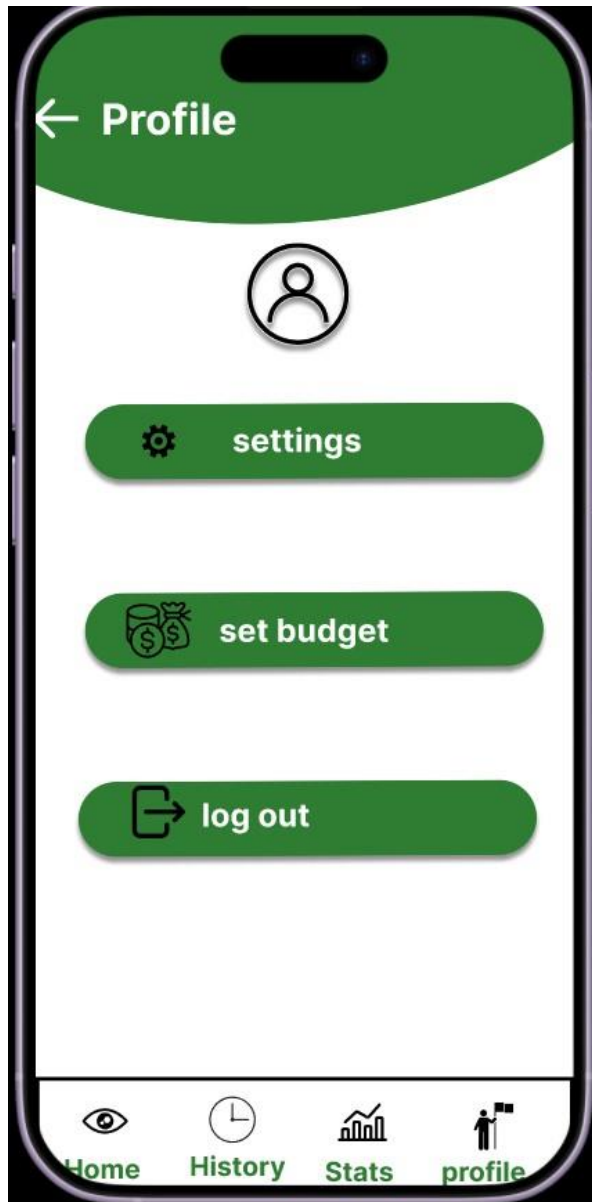
(Source: Figma, 2026)

Statistics Screen:

This screen provides a visual overview of the user's financial data in the Cash Control app. It includes charts such as a pie chart showing expense categories and a bar chart comparing monthly income and expenses. It also displays total income and total expenses, helping users understand their spending patterns (Statista. 2024).

Navigation Steps:

1. From **Home screen**, click **Stats**
2. App opens **Statistics screen**
3. View charts and financial summaries



(Source: Figma, 2026)

Profile Screen:

This screen allows users to manage their account and app settings in the Cash Control app. It provides options such as accessing settings, setting a budget, and logging out. The layout is simple and ensures users can easily control their account and preferences (Statista. 2024).

Navigation:

1. From **Home** screen, click **Profile**
2. App opens **Profile** screen
3. Click **Settings** → Manage app preferences
4. Click **Set Budget** → Define or update budget
5. Click **Log out** → Exit the app

5. PROJECT PLAN

Project Plan

Project phrase	Task	Week 1	Week 2	Week 3	Week 4
Research	Review existing budgeting apps				
Research	Identify key app features				
Planning	Define app requirements				
Planning	Design system structure				
UI Design	Create user interface wireframes				
UI Design	Design navigation flow				
Development	Set up development environment				
Development	Build core app features				
Testing	Test app functionality and fix errors				
Finalisation	Prepare documentation and submit project				

(Source: Figma, 2026)

6. Conclusion

The design of the **Cash Control** app provides a clear plan for developing an effective budgeting tool. Based on research into existing financial apps, it includes key features such as expense tracking, budget limits, alerts, and visual reports. These features support better financial awareness and decision-making (Chen, 2021; Kumar and Lim, 2022).

Overall, this design demonstrates how Cash Control can help users manage their finances more effectively, and it serves as a strong foundation for building the final application prototype (Chen, 2021; Kumar and Lim, 2022).

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